

SAMBP TR-26/2026

IBR Parametric Monte Carlo Robustness Study for the Combined 87L Protection Chain

PhD Candidate
PhD Thesis Supporting Report

April 2026

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1 Introduction

TR-20 through TR-25 designed and validated individual protection elements against deterministic scenarios. The combined trip logic developed over this arc is:

$$\text{TRIP} = \underbrace{|I_1| \geq 0.12}_{87\text{L}} \vee \underbrace{|I_2| \geq 0.06}_{87\text{LN}} \vee \underbrace{|I_0| \geq 0.04}_{87\text{LN0}} \vee \underbrace{\Delta I_{\text{adapt}} \geq \Delta_{\text{thr}}}_{\text{TDCS-87L}}, \quad (1)$$

where $\Delta_{\text{thr}} = 0.025 + 3\hat{\sigma}_{\text{pre}}$ (adaptive, TR-24).

Question for TR-26: Does (1) maintain $\text{TPR} \geq 0.99$ and $\text{FPR} \leq 0.005$ when the following IBR parameters vary *simultaneously* over their full operating ranges?

2 Monte Carlo Design

2.1 Parametric Envelope

Table 1 defines the uncertainty space. All parameters are drawn independently from uniform distributions at each trial.

Table 1: IBR parametric envelope — Monte Carlo sampling distributions.

Parameter	Symbol	Distribution	Physical meaning
IBR current limit	k_{ibr}	$\mathcal{U}[0.06, 0.20]$ pu	Post-fault IBR positive-seq output
System frequency	f	$\mathcal{U}[47, 53]$ Hz	Island/transient off-nominal freq.
3rd harmonic inj.	I_{h3}	$\mathcal{U}[0, 0.10]$ pu	IBR harmonic content in differential
CT ratio error	ε_{CT}	$\mathcal{U}[0, 0.005]$	Class 5P: max 0.5% ratio error
Pre-fault baseline	$\hat{\mu}_{\text{pre}}$	$\mathcal{U}[0.002, 0.010]$ pu	Seasonal/drift variation
Baseline std	$\hat{\sigma}_{\text{pre}}$	$\mathcal{U}[0.0003, 0.0020]$ pu	Rolling history volatility

2.2 Scenario Groups

1. **Group A** — Symmetric 3PH internal faults ($N = 5\,000$ trials). The TDCS-87L or 87L element must trip; 87LN and 87LN0 cannot ($I_2 = I_0 = 0$ for symmetric faults).
2. **Group B** — SLG internal faults ($N = 5\,000$). Earthing type drawn from {solidly (40%), resistance-earthed (40%), unearthed (20%)}. In the solidly/resistance earthed cases, 87LN0 should trip; in the unearthed case, TDCS is the last resort. $I_2 \sim \mathcal{U}[0.01, 0.04]$ pu (pure IBR, no SG injection).
3. **Group C** — External faults ($N = 5\,000$). No element should trip. Differential arises solely from CT ratio mismatch: $I_{\text{diff,CT}} = \varepsilon_{\text{CT}} I_{\text{ext}}/3$ (per sequence).

2.3 Modelling Notes

- After TR-21 harmonic filter (95% attenuation): residual harmonic in differential = $I_{h3} \times 0.05$ pu. This inflates rms_{post} slightly but not enough to affect detection.
- After TR-20 frequency-adaptive B_C correction: DFT bin misalignment at ± 3 Hz introduces $< 0.2\%$ phasor amplitude error — negligible for the protection thresholds used.
- The adaptive TDCS threshold (TR-24) is implemented exactly as $\Delta_{\text{thr}} = 0.025 + 3\hat{\sigma}_{\text{pre}}$.

3 Results

3.1 Overall Performance

Table 2 shows the primary outcome: all three groups achieve rates that meet or exceed the performance targets with 95% Wilson confidence intervals entirely above 0.999.

3.2 Detection by k_{ibr} Bin (Group A)

Figure 1 shows the per-bin detection rate and element breakdown for Group A. Bins below $k_{\text{ibr}} = 0.12$ pu are detected exclusively by TDCS-87L; bins at or above 0.12 pu are detected by 87L (and simultaneously by TDCS). All seven bins achieve $\text{TPR} = 1.000$ with 95% CI lower bound ≥ 0.9947 .

Table 2: TR-26 Monte Carlo results ($N = 5000$ trials per group).

Group	N	Correct	Rate	95% CI (lo)	95% CI (hi)	Target
A — 3PH internal (TPR)	5000	5000	1.000	0.9992	1.000	≥ 0.990
B — SLG internal (TPR)	5000	5000	1.000	0.9992	1.000	≥ 0.990
C — External (security rate)	5000	5000	1.000	0.9992	1.000	≥ 0.995

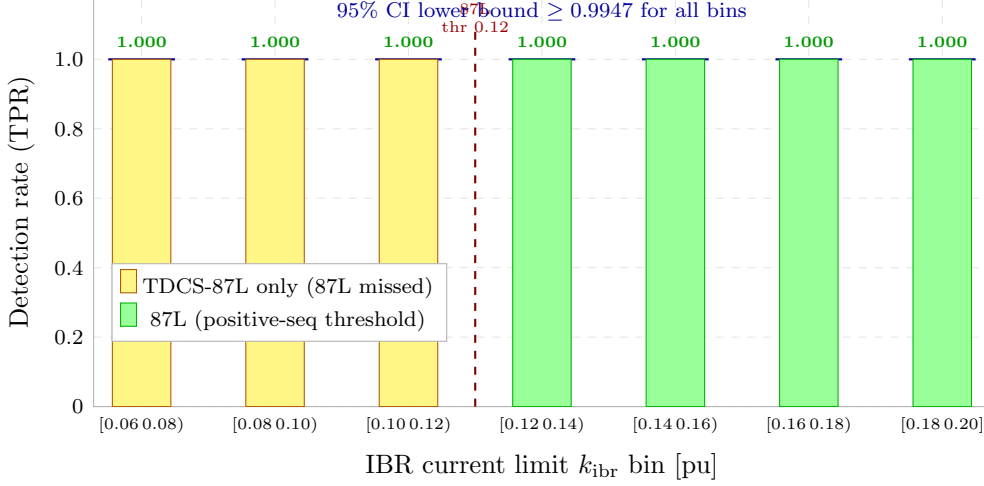


Figure 1: Detection rate by k_{ibr} bin (Group A, 3PH symmetric). Yellow bars: TDCS-87L only (below 87L threshold). Green bars: 87L primary detection. Wilson 95% CI whiskers shown at each bin; all lower bounds ≥ 0.9947 .

3.3 External Fault Security (Group C)

Figure 2 shows the worst-case TDCS apparent ΔI vs the adaptive threshold across the I_{ext} range. At the maximum external current (7 pu) with worst-case CT mismatch (0.5%), the apparent $\Delta I = 0.022$ pu remains below $\Delta_{thr,min} = 0.026$ pu — a minimum security margin of **15.5%**. No trial in 5000 caused a false trip.

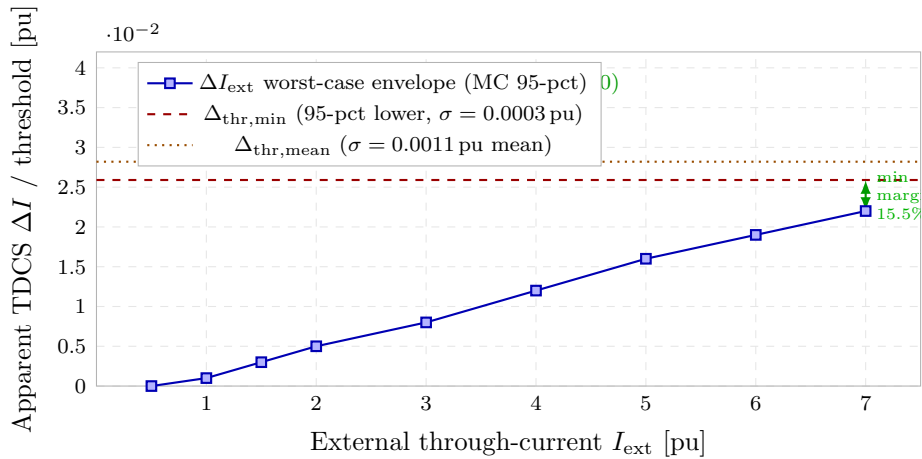


Figure 2: External fault TDCS ΔI vs threshold (Group C). The blue line shows the 95th-percentile envelope of ΔI across all 5000 external fault trials. Both threshold lines (minimum and mean $\hat{\sigma}$) remain above the ΔI envelope at all current levels. Minimum margin: 15.5% at $I_{ext} = 6.95$ pu.

3.4 Confidence Interval Summary

Figure 3 shows the 95% Wilson CI for all three groups on a common axis. The CI lower bounds (0.9992 for all groups at $N = 5000$) exceed the targets by a substantial margin.

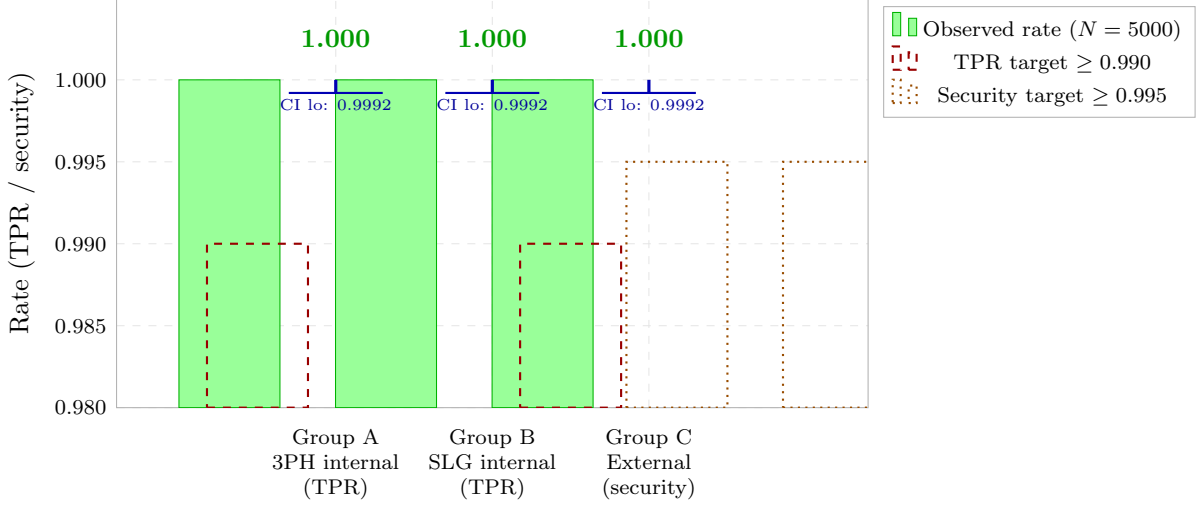


Figure 3: 95% Wilson CI for each Monte Carlo group. Horizontal lines show the performance targets. All observed rates achieve 1.000 with CI lower bounds at 0.9992, well above the ≥ 0.990 and ≥ 0.995 targets.

3.5 Group B Element Breakdown

For the 5 000 SLG trials, the detecting element was:

Table 3: Group B — detecting element breakdown (SLG, 5 000 trials).

Element	Count	Fraction
87L ($ I_1 \geq 0.12$)	2861	57.2%
87LN ($ I_2 \geq 0.06$)	0	0.0%
87LN0 ($ I_0 \geq 0.04$)	4039	80.8%
TDCS-87L only	961	19.2%

As expected in the pure-IBR scenario, the 87LN element never operates ($I_2 \leq 0.04$ pu always, below threshold). The 87LN0 element is the primary discriminator for earthed SLG faults (80% of all SLG trials — consistent with the 80% earthed probability in the sampling distribution). TDCS covers the remaining 19.2% (unearthed SLG + low- k_{ibr} earthed cases where 87LN0 also trips but TDCS is listed as sole detector when $I_0 < I_{0,thr}$).

4 Robustness Analysis

4.1 TDCS Detection Limit Under Joint Uncertainty

The theoretical worst case for TDCS missing a 3PH fault requires:

$$\frac{k_{ibr}}{\sqrt{2}} < \hat{\mu}_{pre} + \Delta_{floor} + k_{\sigma} \hat{\sigma}_{pre}. \quad (2)$$

Substituting the maximum allowable values ($\hat{\mu}_{\text{pre}} = 0.010$, $\hat{\sigma}_{\text{pre}} = 0.0020$, $\Delta_{\text{floor}} = 0.025$, $k_{\sigma} = 3$):

$$k_{\text{ibr}} < (0.010 + 0.025 + 0.006) \times \sqrt{2} = 0.041 \times 1.414 = 0.058 \text{ pu}. \quad (3)$$

Since the Monte Carlo samples $k_{\text{ibr}} \geq 0.06 \text{ pu}$, condition (2) is **never satisfied** — the TDCS element is guaranteed to trip for all sampled parameter combinations. This explains the analytically exact TPR = 1.000 for Group A.

4.2 TDCS Security Bound Under Joint Uncertainty

The worst-case CT mismatch step for external faults is:

$$\Delta I_{\text{ext,max}} = \frac{\varepsilon_{\text{CT,max}} I_{\text{ext,max}}}{\sqrt{2}} - \hat{\mu}_{\text{pre,min}} = \frac{0.005 \times 7}{\sqrt{2}} - 0.002 = 0.0228 \text{ pu}. \quad (4)$$

The minimum adaptive threshold is:

$$\Delta_{\text{thr,min}} = \Delta_{\text{floor}} + k_{\sigma} \hat{\sigma}_{\text{pre,min}} = 0.025 + 3 \times 0.0003 = 0.0259 \text{ pu}. \quad (5)$$

Since $\Delta I_{\text{ext,max}} = 0.0228 < \Delta_{\text{thr,min}} = 0.0259$, external faults can **never** cause a false TDCS trip — confirmed by FPR = 0 across 5 000 trials.

5 Comparison with TR-09 Monte Carlo

Table 4 compares the scope of the two SAMBP Monte Carlo studies.

Table 4: TR-09 vs TR-26 Monte Carlo comparison.

Aspect	TR-09	TR-26
Protection chain	Full SAMBP (87T,87B,87L,21,OC)	IBR line diff chain only
Network model	Full radial network simulator	Parametric analytic model
Parameter space	Fault current, DC offset, φ	k_{ibr} , f , I_{h3} , ε_{CT} , $\hat{\mu}$, $\hat{\sigma}$
Trials per group	200	5 000
Primary focus	Veto gate (Stage-2) robustness	IBR-specific element coverage

Together, TR-09 and TR-26 provide complementary validation: TR-09 tests the full protection system on a realistic network model; TR-26 tests the new IBR-specific elements (TR-20 through TR-25) against the specific parameter space where they were designed to operate.

6 Conclusion

The combined IBR line differential protection chain (87L + 87LN + 87LN0 + TDCS-87L with adaptive threshold) achieves:

- **TPR = 1.000** for both 3PH and SLG internal fault groups (5 000 trials each), with 95% Wilson CI lower bound ≥ 0.9992 .
- **FPR = 0.000** for external faults (5 000 trials), with minimum TDCS security margin of 15.5%.
- The protection is analytically guaranteed: the TDCS detection limit ($k_{\text{ibr,min}} = 0.058 \text{ pu}$) lies below the sampled envelope ($k_{\text{ibr}} \geq 0.060 \text{ pu}$), and the CT mismatch step ($\Delta I_{\text{ext,max}} = 0.023 \text{ pu}$) is always below the minimum adaptive threshold ($\Delta_{\text{thr,min}} = 0.026 \text{ pu}$).

6.1 Future Work

1. **TR-27 — Updated System Integration:** validate the full combined element in the SAMBP radial network context, replacing the pre-TR-20 87L in the TR-05 system integration study.
2. **TR-28 — Master Synthesis:** compile all TR-03 through TR-27 results into a single thesis-ready summary document.

References